Series of Real Numbers - Cheat Sheet

Reference: Series.pdf

1. BASICS & NECESSARY CONDITION

Series: An infinite sum $\sum_{n=1}^{\infty} a_n$.
Sequence of partial sums: $s_n = \sum_{k=1}^{n} a_k$.
If $\langle s_n \rangle$ converges to s, the series converges to s.
Necessary Condition: If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

- Converse is false (e.g., $\sum \frac{1}{n}$ diverges).
- If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ cannot converge.

Non-negative Series: A series of non-negative terms converges iff its partial sums form a bounded sequence.

2. STANDARD SERIES

Geometric Series: $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots$

- Converges if $|x| < 1$
- Diverges if $x \geq 1$
- Oscillates finitely if $x = -1$, infinitely if $x < -1$.

p-series Test (Hyper Harmonic):
$\sum \frac{1}{n^p}$ converges iff $p > 1$ and diverges if $p \leq 1$.
$\sum \frac{1}{n(\log n)^p}$ converges iff $p > 1$.
Telescopic Series: $\sum (b_n - b_{n+1})$.

3. TESTS FOR POSITIVE SERIES

Comparison Test: Let $\sum a_n$ be a positive series.

- If $a_n \leq c_n$ for $n \geq N_0$, and $\sum c_n$ converges $\Rightarrow \sum a_n$ converges.
- If $a_n \geq c_n$ for $n \geq N_0$, and $\sum c_n$ diverges $\Rightarrow \sum a_n$ diverges.

Limit Form Test: Let $\sum a_n, \sum b_n$ be positive.
If $\lim_{n\to\infty} \left(\frac{a_n}{b_n}\right) = l$ (finite, non-zero), then both series converge or diverge together.
Root Test: Let $\alpha = \limsup_{n\to\infty} \sqrt[n]{a_n}$.

- Converges for $\alpha < 1$
- Diverges for $\alpha > 1$
- Test fails for $\alpha = 1$

Ratio Test:

- Converges if $\limsup_{n\to\infty} \left(\frac{a_{n+1}}{a_n}\right) < 1$
- Diverges if $\limsup_{n\to\infty} \left(\frac{a_{n+1}}{a_n}\right) > 1$
- If $= 1$, series may or may not converge.

Rabbe's Test: If $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = l$:

- Converges for $l > 1$
- Diverges for $l < 1$ (fails for $l = 1$)

Cauchy Condensation Test:
If $\langle a_n \rangle$ is a decreasing positive sequence, $\sum a_n$ and $\sum 2^n a_{2^n}$ converge or diverge together.

4. RESULTS ON SERIES

- If $\sum a_n$ converges (positive terms), then $\sum a_n^2$ also converges (converse is false).
- If $\sum a_n^2$ and $\sum b_n^2$ converge, then $\sum a_n b_n$ converges.
- If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum (a_n \pm b_n)$ diverges.

5. ALTERNATING SERIES

Series of the form $\sum (-1)^{n-1} a_n$, where $a_n > 0$.
Leibnitz / Alternating Series Test:
Suppose $\sum a_n$ is an infinite series such that:

(a) $a_1 \geq a_2 \geq a_3 \geq \ldots$
(b) $\lim_{n\to\infty} a_n = 0$

Then the series converges.
Absolute Convergence:
$\sum u_n$ is absolutely convergent if $\sum |u_n|$ is convergent.
Note: Every absolutely convergent series is convergent.
Conditional Convergence:
$\sum a_n$ is conditionally convergent if $\sum a_n$ converges but $\sum |a_n|$ diverges.

6. TESTS FOR ARBITRARY TERMS

Abel's Test: If $\{b_n\}$ is positive monotonically decreasing, and $\sum u_n$ converges, then $\sum b_n u_n$ converges.
Dirichlet's Test: If $\{b_n\}$ is positive monotonically decreasing tending to zero, and $\sum u_n$ has bounded partial sums, then $\sum b_n u_n$ converges.

7. POWER SERIES

Series of the type $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.
Radius (R) & Interval of Convergence:
$\frac{1}{R} = \lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$

- Converges absolutely for $|x| < R$
- Diverges for $|x| > R$
- Interval of convergence is $(-R, R)$.
- Converges uniformly for $|x| \leq \rho < R$.